Socially Responsible Investment and Gender Equality in the United States Census

by

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Abstract

With administrative data, we test whether institutional ownership with a social preference is related to employee-level gender equality. We show that the gender pay gap, which is an unexplained part of the lower wages of female employees, does not have a significant relation with socially responsible investments. Next, we show that female directorship strengthens the relation between socially responsible investments and the gender pay gap. When there are female directors, socially responsible investments have a robust correlation with a lower gender pay gap. This is because female directorship alleviates information asymmetry in gender equality.

Keyword: gender, socially responsible investment, board of directors

JEL Classification: J16, G34, F29

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1. Introduction

Since the early 20th century, most countries worldwide have advocated for equal economic opportunities for both genders. In the United States, Title VII of the Civil Rights Act of 1964 prohibited employment discrimination based on gender, leading to subsequent government regulations promoting equal opportunities. Financial markets have demonstrated a clear preference for gender equality. In the 1970s, several initiatives were created, such as Women's World Banking, to provide funding to female business owners. The demand for gender equality has picked up the pace in the 21st century, and many institutional investors have to include requirements for gender equality in their investment strategies in the 21st Century.[1]

This paper suggests that investor preference for gender equality is weakly correlated with employee-level gender equality. Although a large portion of investors have a clear preference for gender equality at the board level, there is little prior evidence of whether it actually contributed to the gender equality of employees. This is largely due to the lack of employee-employer-matched data for extended time periods. This paper fills this gap by utilizing an updated panel of the United States Census-restricted data. The 2021 snapshot of the Longitudinal Employer-Household Dynamics (LEHD) of the Census has compensation information of employees with demographic information up to 2021, which includes the period with heightened attention for gender equality. Unlike previous literature using survey data or subjective measures of inequality for recent periods in the United States (see, e.g., Sockin and Sockin, 2023; Edmans, Flammer, and Glossner, 2023), we provide a comprehensive and objective measure of gender inequality and study the implications.

[1] Examples include investment companies (Vanguard, State Street, and BlackRock), pension funds (California State Teachers' Retirement System, Japan Government Pension Investment Fund, Swedish Pension Funds, and Australian Super), and banks (BNP Paribas, Standard Chartered, Goldman Sachs, HSBC, PNC banks, and Westpac).

By using this data, we directly test whether the demand of institutional investors has improved the gender equality of employees. Previous papers find a robust relation between board-level gender equality and institutional investment (e.g., Knippen, Shen, and Zhu, 2019; Friedman, 2020). However, we find that the relation does not extend to the employee level, which has a much larger population than the board of directors.[2] More specifically, socially responsible investment is correlated with gender equality, but they are not statistically and economically significant if we control the common variations across time, industries, and regions. We do, however, observe a statistically significant and economically significant relation within a subsample where there is at least one female director in a firm.

Theories have sought to modify the Modigliani-Miller framework to consider investors' non-pecuniary benefits. Researchers such as Hart and Zingales (2017), Edmans, Levit, and Schneemeier (2022), and Dangl, Halling, Yu, and Zechner (2023) have proposed that companies allocate resources to meet the preferences of investors focused on good environmental, social, and governance (ESG) practices. Gender equality is a significant aspect of social practice for portfolio companies.[3] However, two implicit assumptions are required when applying these theories to gender equality: first, that investors have up-to-date workforce information, and second, that managers implement investors' preferences without significant agency problems. However, investors often lack access to detailed demographics and compensation information for general employees. Consequently, they often rely on the composition of the board of directors, which is more readily observable across companies. For example, the big three investment companies (State Street, Vanguard, and BlackRock) implement gender equality by voting against the re-election of

[2] Compustat firms from 2001 to 2021 have 7 directors and 1,232 employees on median.
[3] A survey of Morgan Stanley Institute for Sustainable Investing against asset managers indicate that the issue of the gender diversity is the most important issues among four social agenda (gender diversity, education, health and nutrition, and community development). See Morgan Stanley Institute for Sustainable Investing (2020) for details

board members at companies that failed to increase board diversity.[4] This reliance on observable information highlights the issue of asymmetric information, where investors (principals) depend on managers (agents) to implement workforce preferences. Therefore, it remains an empirical question whether companies' agents effectively implement gender equality practices at the employee-level.

To measure investors' demand for gender equality, we take two different approaches. First, we use the revealed preference of institutional investors by measuring the sensitivity of the institution's holding in response to changes in the female composition of the board of directors within the rolling 5 years of past data. This measure is based on robust findings that institutional investors have a demand for firms with more female directors when they show a preference for diversity (e.g., Knippen, Shen, and Zhu, 2019; Friedman, 2020). The second measure is based on mutual fund names, as in van der Beck (2023). When a mutual fund's name includes a word related to social concerns, we classify them as socially responsible institutional investors. We aggregate socially responsible institutional holdings (or mutual funds' holdings) over outstanding shares within a firm-year to measure socially responsible investment. The gender pay gap is our main dependent variable; it is measured as the wage difference between female and male employees after controlling for their experience, age, education, race, and country of origin (Goldin, Kerr, Olivetti, and Barth, 2017).

Our empirical results suggest that institutional investors' social preference does not necessarily translate into equality of employees. When we regress the gender pay gap on socially responsible investment, the coefficient is not statistically different from zero after controlling for

[4] Although it is a possibility that institutional investors only consider board-level diversity, they specify that their goal is to achieve diversity in general workplace. For example, see a media article "A statue of a defiant girl now faces the Wall Street bull" by Emma Fierberg and Rachael Levy, Business Insider, March 7, 2017.

firm characteristics, year-fixed effects, state-fixed effects, and industry-fixed effects. The 90% confidence interval of the coefficient implies that one percentage point increases in socially responsible investment lead to -0.152 to 0.036 percentage point change in gender pay gap. The results are qualitatively the same when we use the alternative measure of socially responsible investment or when we use simple differences between the wages of female and male employees. We do not specify the causality of the findings. It indicates either that socially responsible investment is not sensitive to employees' gender equality, possibly because of a lack of information, or that socially responsible investment does not improve employees' gender inequality, possibly because of agency problems.

Next, we find that female directors boost the relation between the employee-level gender pay gap and socially responsible investment. The hypothesis comes from the fact that women in higher positions improve gender equality in lower ranks (Cohen, Broschak, and Haveman, 1998; Dezsö and Ross, 2012; Tate and Yang, 2016). Specifically, when we see the relation between the gender pay gap and socially responsible investment in companies with at least one female director (62% of Compustat firms), it is significantly stronger than the ones with all-male director firms (38% of Compustat firms). As a result, socially responsible investment has a statistically and economically significant relation with the gender pay gap if there is at least one female director. Since the existence of female directors is a proxy of omitted variables, we test other hypotheses controlling for effects of female CEOs, sensitivity differences depending on time, and the quadratic relation between socially responsible investment and the gender pay gap. However, the effect of female directors on sensitivity between the gender pay gap and socially responsible investment stays intact.

The effect of the female board of directors on decreasing the gender gap reaches the highest when there are two female boards of directors. The reinforcing effect of two female directors is

2.24 times bigger than the effect of one female director. The effect of at least three directors is not statistically different from the effect of two female directors because it has larger standard errors, diluting the precision of the coefficient.

We provide suggestive evidence that female directors relieve the information asymmetry between firms and insiders. Directors can disseminate information on gender equality and, in turn, decrease the information asymmetry between investors and managers to help reduce discrimination within a company. Prior studies find that directors relieve the information asymmetry between insiders and investors in a context other than gender equality (Fama and Jensen, 1983; Adams, Hermalin and Weisbach, 2010; Armstrong, Core, and Guay, 2014). Motivated by previous findings on directors' roles, we show that when there are more female directors, managers talk more about topics related to gender equality during earnings calls. The results are robust when we include various control variables and fixed effects or when we change the list of gender-related words. The results imply that the existence of female directors amplifies the propagation of gender-related information to investors.

Our papers contribute to the literature by providing evidence of the real effects of socially responsible investment. Many papers argue that ESG investment has real effects, such as investment, labor decisions, and carbon emissions (e.g., Heinkel, Kraus, and Zechnerm, 2001; Hartzmark, and Shue, 2022; De Angelis, Tankov, and Zerbib, 2023; Dangl et al., 2023; Li, Shan, Tang, and Yao, 2024). Other papers point out the informational or technological obstacles that prevent the real effects of ESG investing (Ilhan, Kureger, Sautner, and Starks, 2023; Yin, 2024). Although some papers looked into the effects of socially responsible investment on leadership-level gender equality (Friedman, 2020), no papers have looked into the relation with employee-

level gender equality, mainly due to the availability of administrative data. Our paper is expected to fill this gap by providing micro evidence of the real impact of socially responsible investment on gender equality.

We also contribute to the literature related to determinants of gender equality. The gender gap in wages, hiring, and promotions has received considerable attention in academic and policy discourse (e.g., Altonji and Blank, 1999; Goldin, 2014; Blau and Kahn, 2017). Papers show that gender equality in a firm is correlated to wage information disclosure (Bennedsen, Simintzi, Tsoutsoura, and Wolfenzon, 2022), availability of negotiation (Card, Cardoso, and Kline, 2016), and female leadership (Tate and Yang, 2015). We test whether socially responsible investment can contribute to improving gender equality in a firm. We also identify the frictions, such as agency conflicts, that investors' preference does not translate into employee-level gender equality.

Our analysis proceeds as follows. In Section 2, we describe the data and how we construct measures of gender equality and socially responsible investment. In Section 3, we show how gender equality is unconditionally related to socially responsible investment. In Section 4, we show female directors strengthen the relation between gender equality and socially responsible investment. In Section 5, we show that female directors alleviate the information asymmetry on gender equality between insiders and investors. We conclude in Section 6.

2. Data and measures

2.1. Data

We construct a longitudinal dataset of public companies from 2001 and 2021 in the 2021 snapshot of the US Census Bureau's Longitudinal Employer-Household Dynamics (LEHD).

LEHD data are collected from the unemployment insurance records of states participating in the program. Missing employment is different slightly from state to state due to variations in state unemployment laws, but the data generally exclude agricultural employees, federal employees, domestic workers, nonprofit employers, and religious organizations. LEHD does not update the data on a yearly basis, and the previous snapshot was in 2014. The updated snapshot covers periods upto early 2020s, when socially responsible investment has shown significant growth (Gibson Brandon et al., 2022). Therefore, the use of the updated LEHD is critical to investigate the relation between gender inequality and socially responsible investments. The starting year of analysis is 2001 because eight states started to be in LEHD data in early 2000, and the socially responsible investment variable is available from 2001.[5] Researchers not affiliated with the US Census Bureau have access to the 26 states.[6] Some companies may have headquarters in non-sample states (for example, in Florida) and have establishments in LEHD states (for example, California), and they are included in the sample as long as they pass the filing criteria described below.

LEHD data are merged with Compustat at the firm-year level using Compustat–SSEL Bridge, and they are further merged with BoardEx and CRSP. After merging with Compustat, BoardEx, and CRSP, we filter the data to US-incorporated firms (Compustat or CRSP acronym: FIC) whose common stocks (10≤SHRCD≤11) are listed in NYSE, AMEX, and Nasdaq (1≤EXCHCD≤3). Also, firms need to have at least 10 employees to be in our samples after removing the top seven high-income employees. Because we do not directly observe the occupation of the individuals and the median number of board of directors during the sample period is seven, we exclude seven high-

[5] BoardEx data starts from 2000. As section 2.3. elaborates, we use the past 5 years to identify gender-sensitive institutional investors. Therefore, the analysis starts from 2001.

[6] They are Arizona, California, Colorado, Connecticut, Delaware, Indiana, Iowa, Maine, Maryland, Massachusetts, Nevada, New Jersey, New Mexico, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, South Carolina, South Dakota, Tennessee, Utah, Virginia, Washington, Wisconsin, and Wyoming.

income employees to focus on the employee-level gender pay gap. We further restrict it to those with at least one board of directors on BoardEx data and with valid institutional ownership data of Thomson/Refinitiv 13F Holding data. The resulting number of observations is 54,000 firm-years, which is rounded to the nearest 500 observations for the observance of the US Census disclosure requirement. After filtering, we combine the data with the CRSP Mutual Fund database.

2.2. Measures: Gender pay gap

We measure the gender pay gap by looking into abnormal salary differences after controlling for individual characteristics within a firm-year as in Goldin et al. (2017). It can measure the unexplained part of wage differences between male and female employees after controlling for age, experience, race, education levels, and country of origin. Specifically, for a firm-year, we construct a panel with all employees except for the top seven high-income employees. In this way, we can more accurately measure the employee-level gender pay gap, rather than the director level. Then, we regress the logarithm of annual wage on an indicator variable for gender along with other individual characteristics within each firm-year.[7] The definition of other individual characteristics is elaborated in Appendix Table A1.

$$\log(AnnualWage)_{i,t} = \beta_{f,t} * 1(Female)_i + Race_i + BirthCountry_i + Education_i + Experience_{i,t} + Age_{i,t} \quad (1)$$

[7] LEHD classifies gender only based on gender-at-birth, rather than individuals' sexual identities by year. In this paper, we use the term "gender" to represent gender-at-birth for the sake of brevity.

Here, coefficient $\beta_{f,t}$ represents the percentage difference of female wage compared to male workers' wage in a firm *f* in year *t*. We take the negative value of $\beta_{f,t}$ and define it as the gender pay gap. Therefore, the higher gender pay gap implies female employees are more disadvantaged compared to male employees. Since the dependent variable is the logarithm of annual wage, minus $\beta_{f,t}$ is percentage discount on salary of female employees.

Figure 1 depicts the kernel density plot of the gender pay gap. To avoid the risks of exposing the identity of individual companies as required by the Census Bureau, we use a kernel density plot instead of a histogram and truncate at the 5% and the 95% level. The figure indicates that the gender pay gap of all firm-years after truncation lies in positive regions, meaning females are disadvantaged in most companies across years. The average gender pay gap is 24.8%, which is close to the prior literature.[8] It means that female employees earn 0.752 dollars when male counterparts earn 1 dollar of wage. The distribution has high variations, whose standard deviation is 15.5%. This implies that some firms have almost zero gender pay gaps and others have large gender pay gaps.

Alternatively, we do not rely on the parametric assumptions of equation (1). We take a simple approach, subtracting the mean (median) female workers' wages from the mean (median) male workers' wages. We scale it with the mean (median) wages of all workers. The average gender pay gap is 28.8% for the difference in mean and 29.3% for the difference in median. They are higher than the gender pay gap estimated with equation (1), and the difference is statistically significant at the 1% level. A higher gender pay gap without controlling characteristics implies that social

[8] Previous papers find that the gender pay gap is 24.7% from 2009 to 2011 in the United States (Goldin, 2014), 21.2% from 1993 to 2019 in the United Kingdom (Lagaras, Marchica, Simintzi, and Tsoutsoura, 2022), and 15.4% from 2003 to 2005 in Denmark (Bennedsen et al., 2022)

discriminations outside of firms result in undesirable characteristics for females, such as lower education levels and shorter experience.

While the gender pay gap can be interpreted as the quality aspect of gender inequality, we also examine the quantity aspect by looking into the proportions of female workers in a company. Specifically, we take a look at how many of the employees are female within a firm-year. Similar to the gender pay gap, we exclude the top seven high-income individuals within each firm-year. During the entire sample period, the proportion is 42.2%, implying that there are fewer females than males in a firm. The number is consistent with the statistics from the Bureau of Labor Statistics, showing that the female labor participation rate (56.5%) is lower than the male counterpart (67.8%) in 2021.

Figure 2 describes the time series of the gender pay gap and the gender proportion of employees. Panel A shows that the gender pay gap has decreased steadily over the last two decades. The results are consistent for different measures of the gender pay gap. It was 28% in the year 2001, and it decreased to 21% in the year 2021. When we use the difference in mean (median), it decreases from 34% (33%) to 24% (24%).[9] The steady decrease in the gender pay gap indicates that the social drive for gender equality has paid off over the last two decades. The proportion of female employees decreases slightly from 43% in 2001 to 41% in 2021. It is consistent with previous literature (Olivetti and Petrongolo, 2016) and policy reports that female participation rates decreased after the Global Financial Crisis and the COVID-19 crisis.[10] However, the proportion of female board directors dramatically increased from 9% to 28%.

[9] The difference between the residual measure and the two simple measures decreases over time from 7% to 3%, and it implies that the disadvantages in characteristics that women had decrease over time.
[10] For further discussions, see the Bureau of Labor Statistics Report 1103, Apr 2023.

2.3. Measures: Socially responsible investments

For the main explanatory variables, we utilize two separate methods to identify socially responsible investments. Firstly, we use the revealed preference approach by measuring the sensitivity of institutional holdings in response to changes in the female composition of the board of directors with the previous 5 years' data. Institutional investors usually use the proportion of female directors to gauge gender equality within a company. For example, State Street Global Advisor, Vanguard, and BlackRock, which are the three biggest investment companies, use a number of female directors to measure gender equality. Prior literature also suggests that institutional investors have a demand for firms with more female directors (Knippen, Shen, and Zhu, 2019; Friedman, 2020). By looking into institutional investors' trades when firms appoint female directors, we identify institutions which reveal their preference for gender equality.

Specifically, for each institutional investor *i*, we analyze how they change their holdings depending on the number of female directors for the previous 5 years.

$$\Delta \frac{Investment}{AssetsUnderManagement}_{i,f,t-n} = \gamma_{i,t} * \Delta \frac{FemaleDirectors}{AllDirectors}_{f,t-n} + Controls_{f,t-n} \ (1 \leq \text{n} \leq 5) \quad (2)$$

Controls include firm size, past returns, and profitability. If an institutional investor adjusts their holdings to a higher proportion of female directors, the $\gamma_{i,t}$ has a positive value. If an institution-year has positive $\gamma_{i,t}$ and it is statistically significant, we classify the institution as socially responsible in year *t*. For a firm's socially responsible investment, we aggregate holdings by socially responsible institutions and scale them by the number of shares outstanding.[11] This

[11] It measures ownership rather than investment, i.e. changes in ownership. We still name it as socially responsible investment, since it is commonly used words in literature and media.

approach has the advantage that it can capture the general demands for social concerns regardless of institutions' types because 13F includes various institutional investors, such as mutual funds, hedge funds, and pension funds. However, it may capture other demands if the nomination of a female director and institutional investment are jointly correlated with omitted demands for stocks. Also, it may capture the potential demands of institutional investors who have a preference for board-level gender equality but not necessarily for employee-level gender equality.

To complement the aforementioned drawbacks of the first measure, we use mutual fund names, as outlined in van der Beck (2023), that include words related to social concerns to classify them as socially responsible institutional investors. We aggregate socially responsible mutual funds' holdings within a firm-year and divide them by the number of shares outstanding.[12] This approach clearly identifies demands for social concerns, because the mutual funds with social agenda on their names have a clear preference for gender equality. However, it significantly underestimates the entire demand for social concerns because it does not include other institutional investors than mutual funds and also because mutual funds are known to incorporate social factors in their prospectus even though they are not focused on ESG funds (Nofsinger, Sulaeman, and Varma, 2019). In summary, the second approach has a lower chance of type-1 errors at the cost of a higher chance of type-2 errors than the first approach. We use both approaches to complement each other.

Table 1 describes the summary statistics of the main variables and control variables. Since ranking variables (such as median and maximum) are subject to higher disclosure requirements by the US Census Bureau, we report mean, standard deviations, and skewness. All continuous

[12] van der Beck (2023) measures the mutual funds' demand for environmental, social, and/or governance factors. We start with his list of words. However, because the gender equality is only clearly related to the social factor, we do not include words that are related to only environmental or governance concern. After filtering them, the word list we use is: social, impact, fair, gender, screen, ethical, conscious, SRI, ESG, and CSR. The last three words are case-sensitive, and other words are not case-sensitive.

variables are winsorized at the 1% and the 99% levels. The definition of control variables is elaborated in Appendix Table A1. As shown in Figure 1, the gender pay gap is positive and subject to large standard deviations. The positive skewness of the pay gap also indicates some firm-years have a very large gender gap. When we control for characteristics, the standard deviation is lower, and the skewness is closer to zero than measured based on the mean or median. This implies that a large portion of variations in the gender pay gap arise from variations in pre-existing discrimination against female employees, such as education.

Socially responsible investment is 0.65% of outstanding shares when we use the revealed preference approach and 0.03% when we use mutual fund names. The number is significantly smaller than media reports on ESG investing.[13] It is because we only focus on social factors and other measures rely on self-reporting of institutional investors, which is subject to self-reporting biases (Boffo and Patalano, 2020; Chen, Cohen, and Gurun, 2021; Gibson Brandon et al., 2021). We rely on investors' actions, such as adjustments of ownership and naming of funds, so they account for lower ownership of a company.

In the last column of Table 1, we show the averages of Compustat companies. We restrict Compustat firms to ones incorporated in the United States, whose common stocks are traded in NYSE, AMEX, and Nasdaq, have at least ten employees, and non-missing observations in Thomson/Refinitiv 13F Holding data. Because there is no data on employees' gender and wage in Compustat, we do not report the gender equality variables. The firm characteristics of sample firms are comparable to the average Compustat firms. If all employees are hired in non-LEHD states, they are out of our samples, so the number of our sample firms is about 6.3% less than the Compustat firm-years during the sample periods.

[13] For example, Gibson Brandon et al. (2022) report that 38.4% of institutional holdings are by institutional investors that signed UN Principles for Responsible Investment (PRI) in North America in 2017.

3. Relation between socially responsible investment and the gender pay gap

We employ a standard empirical methodology by regressing the gender pay gap on lagged socially responsible investment with a firm-year panel. Although we use the lagged dependent variables to alleviate endogeneity, the results are qualitatively similar when we examine the contemporaneous relation. We include control variables related to labor costs, such as firm size, profitability, and investment (e.g. Brown and Medoff, 1989; Card, Devicienti, and Maida, 2014). Depending on columns, we include various fixed effects such as industry-fixed effects (3-digit Standard Industrial Classification (SIC)), state-fixed effects, and year-fixed effects.[14] Alternatively, we include industry-times-year fixed effects or state-times-year fixed effects, which account for time-varying industry shocks or regional shocks to the level of the gender pay gap. Because both the dependent variable and the explanatory variables are correlated within a firm, we cluster the standard errors by a firm.

Table 2 shows the relation between the gender pay gap and socially responsible investment. The relation is negative in every specification, but the statistical and economic significance greatly differ depending on the inclusion of fixed effects. Column 1 shows the coefficient on socially responsible investment is -0.481, and it is statistically significant at the 1% level. This means that when socially responsible investment is the sole explanatory variable, it is associated with a smaller gender pay gap. If a firm has 1% more socially responsible investment, which equals to

[14] Firms can have multiple establishments in different states for each year. In this case, we set the establishment with the most employees as their location of the state-fixed effects.

0.73 standard deviation, the gender pay gap in a firm is 0.48% smaller. The unconditional average of the gender pay gap is 24.8% as in Table 1, so it is 1.9% lower than the unconditional average. Adding firm characteristics in Column 2 does not greatly affect the relation of interests, although all the control variables have a significant relation with the gender pay gap. However, it loses statistical and economic significance when we include fixed effects. Column 3 includes industry-, state-, and year fixed effects, and Column 4 includes industry-times-year fixed effects and states-times-year fixed effects. The magnitude of the coefficient on socially responsible investment decreases to -0.058 in Column 4, and it is not statistically significantly different from zero (p-value equals 30.8%). The difference between Columns 1 to 2 and Columns 3 to 4 comes from the secular decreases in the gender pay gap over the last two decades. As Figure 2 indicates, the gender pay gap has decreased by 7% over the last two decades, and it coincides with the trend that socially responsible investment has increased. The socially responsible investment was 0.75% since 2010 compared to 0.44% before 2010. Higher socially responsible investment may not be the only reason why the gender pay gap lowers in post-2010.

Turning attention to the control variables, we find a robust positive relation between the gender pay gap and firm age or firm size. Previous papers point out that male employees benefit more when the firm size is bigger, possibly because of career interruption for female employees due to motherhood (Oi and Idson, 1999; Bertrand, Goldin, and Katz, 2010) The effects of leverage flip signs depend on the inclusion of fixed effects.

Column 5 uses an alternative measure of gender pay gap from the difference in mean, and Column 6 relies on the difference in the median. The results are qualitatively the same as Column 4, meaning that the effects of socially responsible investment are not significantly different from

zero. In Column 7, we measure the socially responsible investment with the name of mutual funds and reach the same conclusion.

Although we are not able to reject the null hypothesis that there is no relation between the gender pay gap and socially responsible investment, neither can we reject the alternative hypothesis that there is a negative relation between the gender pay gap and socially responsible investment. First, the 90% confidence interval of the coefficient on socially responsible investment ranges from -0.152 to 0.036 (Column 4 of Table 2). It implies that the true relation is still more likely to be negative than positive. Second, it is possible that there is noise in measuring socially responsible investment. Although we use two different measures of socially responsible investment, they are not a perfect proxy of socially responsible investment. It is not clear whether the noises in independent variables strengthen or weaken the relations, but it clearly enlarges the standard errors of the coefficient. Finally, although the year-fixed effects decrease the significance of socially responsible investment, at least a portion of year-fixed effects reflect the impact of socially responsible investment. The secular decrease in the gender pay gap can arise from the steady increases in socially responsible investment. For these reasons, we do not argue that our results are evidence of the ineffectiveness of socially responsible investment.

In Appendix Table A2, we redo the analysis in Table 2 but replace the socially responsible investment with all institutional investment or non-socially responsible investment. All institutional investment is defined as sum of institutional holdings in 13F over outstanding shares, and non-socially responsible investment is all institutional investment minus socially responsible investment. There is a positive and significant relation between the gender pay gap and all institutional investment (or non-socially responsible investment), implying that socially

responsible investment in Table 2 is at least more sensitive to the gender pay gap than general institutional investment.

Next, we replace the gender pay gap with the proportion of female employees in a firm. Table 3 presents the results indicating that socially responsible investment is weakly correlated with fewer female workers in a company. The relation is statistically significant at the 10% level when we include various fixed effects in Columns 2 and 3. A negative relation indicates that socially responsible investment is at least not positively correlated with how many female workers are in a company. It is strikingly different from the effects on female directors of a board. Columns 3 to 6 indicate that socially responsible investment is positively correlated with the proportion of female directors, and it is statistically significant at the 5% level when we include fixed effects. It is consistent with the prior studies that socially responsible investment is correlated with director-level gender equality. Our results indicate that the argument cannot be directly extended to employees.

The results in Tables 2 and 3 show that, although socially responsible investments are associated with a higher proportion of female directors, it does not have a significant relation with the gender pay gap. This discrepancy arises from different levels of information availability for directors and employees. The proportion of directors is readily available to all investors through disclosures, but the proportion and wage information are not available to investors.

Furthermore, we test the implications of information availability related to gender equality on the market valuations of a company. When investors prefer firms with a lower gender pay gap, we

can expect that those firms have higher market valuations after controlling for firm characteristics. However, if the employee-level gender pay gap is not communicated to the investors, the gender pay gap can have an insignificant relation with the market valuations.

Specifically, in Table 4, we regress the year-end market value of assets over the book value of assets on the current year's employee-level gender pay gap and the proportion of female directors. We include control variables about firm size, firm age, and profitability to control for the determinants of market valuations. We also include industry-times-year fixed effects and state-times-year fixed effects.

Column 1 in Table 4 indicates that the employee-level gender pay gap is not significantly related to the market valuations. Although not significant, the sign of the coefficient is negative. It is starkly different from the results with the proportion of female board of directors in Column 2. One percentage point increase in the proportion of female directors is related to a 0.59 percentage point increase in market valuations, which are economically and statistically significant. In Column 3, we include both the gender pay gap and the proportion of female directors. Surprisingly, the coefficient on the gender pay gap is *negative* and statistically significant at the 10% level. The results show that although director-level gender equality gathers the demand of investors, the employee-level gender pay gap does not lead to higher market valuations.

To summarize, socially responsible investment does not have a significant relation with the gender pay gap or the proportion of female employees. In the following section, we show that the relation is stronger for a subsample where it has at least one female board of directors, and we also investigate the reasons for different sensitivity.

4. Sensitivities depending on female board of directors

4.1. Interaction between socially responsible investment and female directors

The drive for gender equality has been prominent on the board of directors. Many institutional investors voluntarily implement explicit or implicit investment mandates that they will not invest in a company without a female board of directors (e.g. State Street, BlackRock, and California State Teachers' Retirement System). Thanks to this social drive, the proportion of female directors increased from 9% in 2001 to 28% in 2021. In this section, we investigate whether the presence of a female board of directors affects the relation between socially responsible investment and the employee-level gender pay gap.

The presence of female directors can strengthen the relation in two ways. First, the female board of directors can convey information about gender equality to outside investors, meaning less information asymmetry. Second, female directors are more likely to implement equal pay to employees when there are institutional demands for it. Previous studies, such as Tate and Yang (2016), find that female leadership leads to a lower gender pay gap, implying a less agency problem for female leadership on the topic of gender equality.

To test the effects of female directors on the relation between socially responsible investment and the gender pay gap, we include an interaction variable between socially responsible investment and an indicator variable of female directors. This indicator equals one when a firm-year has at least one female board of directors at the end of the year. The unconditional average of this indicator variable of the Compustat company is 61.8%.

Table 4 presents the results with the interaction variables. Similar to previous tables, we present the results without fixed effects and with various fixed effects. Unlike the coefficient on the sole variable of socially responsible investment, the coefficient on the interaction between socially responsible investment and female directors is statistically significantly negative with multiple fixed effects. In Column 1, the coefficient is 0.333, and it implies that socially responsible investment is correlated with less gender pay gap when there is at least one female director. In Columns 2, we include industry-, state-, and year-fixed effects. The coefficient of the interaction term is 0.271, which is statistically significant. The result implies that the effects of socially responsible investment are stronger than when there are no female directors. Note that the insignificant positive coefficient on the sole variable of socially responsible investment is offset by the negative coefficient on the existence of female directors because they have a positive correlation coefficient (i.e., firms with higher socially responsible investment are likely to have at least one female directors, as shown in Table 3). Therefore, when we restrict our samples to ones with at least one female director, the relation between the gender pay gap and socially responsible investment is significantly negative. The results are comparable when we control time-varying industry-fixed effects and state-fixed effects in Column 3. Also, the results are robust when we replace the dependent variables with simpler measures in Columns 4 and 5, although it is statistically significant only at the 10% level when we use the difference in mean wage. When we use the socially responsible investment with mutual fund names, the interaction term is statistically significant. The magnitude is much higher than when we use the measures based on revealed preference, and it implies that the effects are larger when we use less noisy measures of socially responsible investment. The last column replaces the gender pay gap with the proportion of female workers. The interaction term is not significant, and it implies that the results of the gender pay

gap are not coming from upscaling or downscaling employees but from wage redistribution among employees.

It is also noteworthy that the coefficient on the female director indicator variable loses statistical significance when we include time-varying industry-fixed effects and state-fixed effects in Columns 3 to 5. It implies that female directors significantly alleviate the gender pay gap only if it is accompanied by socially responsible investment. Taken together, the lower gender pay gap necessitates at least one female director as well as higher socially responsible investment.[15]

In Appendix Table A3, we replace the gender pay gap with the proportion of female employees. Unlike the results on the gender pay gap in Table 5, the interaction between socially responsible investment and female directors does not have statistical or economic significance. This result implies that socially responsible investment is not related to the quantity aspects of gender inequality no matter there are female directors or not.

4.2. Alternative explanations

The previous subsection 4.1. shows the robust coefficient on the interaction between socially responsible investment and the female director indicator. However, the significant coefficient on the interaction term may arise from the omitted variable that is correlated with the existence of female directors.

First, firms with at least one female director are more likely to have a female CEO, and the female CEO may explain the significance of previous results. In Column 1 of Table 6, we replace

[15] In Column 3 of Appendix Table A2, we show that the gender pay gap has negative relationship with the proportion of female directors. The result in Table 5 indicates that this negative relationship is driven by firms with higher socially responsible investment.

the female director indicator with the female CEO indicator. The sole variable of the female CEO indicator is significantly correlated with less gender pay gap, consistent with previous literature that female managers alleviate gender pay gap (Tate and Yang, 2016). However, the interaction between socially responsible investment and the female CEO indicator is not statistically different from zero, which is clearly different from the effects of female directors. Together, it implies that female CEO alleviates the gender pay gap, but it does not depend on the level of socially responsible investment. It is a stark difference from the case of female directors, as Column 3 in Table 5 indicates that female directors are effective when coupled with socially responsible investment.

Second, the existence of a female director is not linked to a positive relation between non-socially responsible investment and the gender pay gap. In Column 2, we define non-socially responsible investment by subtracting socially responsible investment from the entire institutional ownership identified in 13F. Although the interaction between non-socially responsible investment and the existence of a female director is positive marginally significant at the 10% level, its magnitude (-0.016) is much less than the interaction between socially responsible investment and the existence of a female director (-0.266 of Column 3 in Table 5). It is also less than the magnitude of the coefficient on sole non-socially responsible investment (0.022). It implies that a firm with a female director does not have a negative relation between non-socially responsible investment and the gender pay gap.

It is possible that the significant coefficient on the interaction between female directors and socially responsible investment may come from the non-linear relation between socially responsible investment and the gender pay gap. In other words, the gender pay gap may have a negative relation with socially responsible investment only if it is high enough socially responsible

investment. If the existence of the board of directors is a proxy of higher socially responsible investment, the coefficient of the interaction term may come from the non-linear effects of socially responsible investment. In Column 3 in Table 6, we address this concern by including the squared term of socially responsible investment as an additional control variable. The interaction term stays highly significant. More importantly, the magnitude of the coefficient (0.265) does not change much from the benchmark result (0.266) in Table 5. The coefficient on the squared socially responsible investment is zero, implying that the relation between socially responsible ownership and the gender pay gap is not quadratic. Altogether, Column 2 indicates the effect of female directors is not from the exponential effect of socially responsible investment.

Finally, the existence of female directors may proxy the effects in the later part of the sample. van der Beck (2023) argues that the flow to ESG-friendly funds picked up the pace since 2012. 48.4% of firms had at least one female director before 2012. The number increased to 74.0% since 2012. Therefore, the significance of the interaction term may indicate that socially responsible investment had a bigger effect in post-2012, where ESG investing earned popularity. However, we showed that the significance of the interaction term stays highly significant when we include year-fixed effects in Table 5. Additionally, we include the interaction of the indicator variable for post-2012 years and socially responsible investment in Column 4 of Table 6. The interaction between socially responsible investment and female directors stays significant, and the magnitude does not change much. The result indicates that the effect of the board of directors does not arise from the difference in sample periods.

In summary, we show that the effect of female directors on the relation between socially responsible investment and the gender pay gap does not arise from an omitted variable that is correlated with female directors.

4.3. Sensitivities depending on the number of female directors

We show that the relation between socially responsible investment and gender pay gap becomes significant when there is at least one female director. A natural question is how many female directors should be in a firm to have a stronger relation between socially responsible investment and less pay gap.

In this subsection, we dissect the indicator variable of female directors into multiple indicators depending on the number of female directors. Specifically, we include indicator variables for one female director, two female directors, and at least three female directors. Then, we interact each indicator variable with the socially responsible investment. Table 7 presents the result. The coefficient on the interaction between socially responsible investment and "1(Female Director = *n*)" indicates how much the effect of socially responsible investment differs when the female director is *n* compared to zero female directors. The interaction between two female directors and socially responsible investment has the biggest coefficient, and it is statistically significant at the 1% level regardless of which fixed effects we use. On the contrary, the interaction between one female director and socially responsible investment is still positive but loses statistical significance when we include industry-, state-, and year-fixed effects. The interaction between at least three female directors and socially responsible investment is not statistically significant at the 5% level, but the coefficient is negative, and it is not statistically different from the interaction between two female directors and socially responsible investment.

The results of the exercise indicate that two female directors have the most significant reinforcing effect on the sensitivity between the gender pay gap and socially responsible investment.

5. Information asymmetry on gender information and female directors

This section investigates why the existence of female directors promotes the relation between socially responsible investment and less gender gap. We propose that female directors alleviate the information asymmetries on gender information between insiders and investors.

Assuming that socially responsible investment has a clear preference for gender equality of employees, information asymmetry between principals (institutional investors) and agents (managers) causes friction in implementing gender equality of employees. If the information is not perfect, managers' actions may be inconsistent with investors' preferences. Also, imperfect information may disrupt institutional investors from identifying companies with good gender equality.

Female directors can alleviate the information asymmetry on gender equality. Prior studies indicate that females in higher ranks can alleviate the discrimination of the lower-rank labor force (Cohen, Broschak, and Haveman, 1998; Dezsö and Ross, 2012; Tate and Yang, 2016). For example, Tate and Yang (2016) find that income loss from plant closure is more serious for female workers, but the effect is alleviated when there are female leaders. Directors can disseminate information on gender equality and, in turn, decrease the information asymmetry between investors and managers to help reduce discrimination within a company. Prior studies find that directors relieve the information asymmetry between insiders and investors in a context other than

gender equality (Fama and Jensen, 1983; Adams, Hermalin and Weisbach, 2010; Armstrong, Core, and Guay, 2014).

We test whether companies with female directors have better information sharing on gender-related information than companies without female directors. To do so, we test whether managers mention gender issues more often in earnings calls when there are more female directors. The motivation behind this analysis is that managers are more eager to express gender-related information when there is pressure from female directors.

Specifically, we combine earnings calls from Seeking Alphas, financial information from Compustat, and board compositions from BoardEx.[16] The number of Compustat firm-years with at least one valid director data in BoardEx is 63,215 from 2006 to 2021. After filtering firm-years without earnings calls and gender-related industries (apparel and cosmetic industries), it becomes 38,153. We use a simple bag-of-words method to examine whether firms with more female directors mention more words related to gender equality. We use two groups of word lists. The first wordlist includes "gender," "female," "DEI," and "D&I," and the second wordlist is the first-word list plus "diversity," "inclusion," and "equality." We separate the word list because the latter three may include topics unrelated to gender equality of employees (such as "product diversity"). To remove the effects of words related to products, we exclude firms in apparel industries ($2300 \leq SIC < 2400$ or $SIC = 5621$) and cosmetic industries ($SIC = 2844$ or 7231). The dependent variable of this analysis is the proportion of gender equality words, which is the number of words related to gender equality mentioned in earnings calls throughout the year, scaled by the total

[16] For this analysis, we use different samples from previous analyses. It is because the US Census Bureau recommends researchers to conduct analyses with public datasets, if the analysis does not require usage of restricted data. This section of analysis does not directly include employees' wage information, so we conduct the analysis with public datasets. As Table 1 indicates, the firm characteristics of Compustat companies are comparable to the samples used in previous analyses.

number of words spoken in earnings calls. We use managers' scripts, including both introductory presentations and answers to the Q&A sessions. We aggregate quarterly earnings calls within a year to increase the power of the analysis. It is missing when there are no earnings calls in a year. We regress the proportion of gender equality words on the previous year-end female directors. To be consistent with the main analysis, we use the indicator variable for at least one female director. As control variables, we use the same set of control variables used in previous tables. Since fixed effects account for large variations in previous analyses, we also include different sets of fixed effects in this analysis.[17]

Table 8 shows the results of the analysis. Columns 1 to 4 use the proportion of the first wordlist as dependent variables. Column 1 indicates that there is a statistically significant relation between the female board of directors and the proportion of gender-related words in earnings calls. The coefficient of 0.765 indicates that companies speak 22.1% more than the unconditional average (0.04% of total words in the script) when there is at least one female director. The magnitude decreases as we include control variables and granular fixed effects. Column 4 shows the coefficient for female directors is 0.155. It indicates that companies speak 4.5% more than the unconditional average when there is at least one female director. Columns 5 to 8 use the second wordlist word proportions as the dependent variables. The statistical and economic magnitudes of female directors are stronger than when we use the first wordlist. Note that it is not a mechanical result that we have more words on the list because the coefficient captures the covariance between the proportion of words and female directors. The coefficient for female directors is 13.007 in Column 5 and 2.392 in Column 8. This means that when there are female directors, firms talk

[17] Unlike the analyses with Census restricted data, we use the state information on Compustat. They are most recent location of the headquarter, so it captures the different location if firms move their headquarters during the sample periods.

about gender-related topics 32.4% and 6.0% more than the unconditional average (0.40% of total words in the script), respectively.

In summary, this section provides suggestive evidence that female directors relieve information between insiders and investors.

6. Conclusions

We document the insignificant relation between socially responsible investment and the employee-level gender pay gap after controlling for unobservable covariance within industries, regions, and time. The 90% confidence interval of the coefficient indicates that one percent more socially responsible investment is related to -0.152% to 0.036% percentage point change in gender pay gap in a firm. The relation is robust to the different measures of gender pay gap and socially responsible investment. Also, socially responsible investment is not positively correlated with the proportion of female workers in a company.

We uncover that the existence of female directors reinforces the relation between socially responsible investment and the gender pay gap. The 90% confidence interval indicates that one percentage point increases in socially responsible investment in a company with female directors is related to -0.458% to -0.074% percentage point change in gender pay gap. The results are not driven by the time trends of female directors, manager effects, or quadratic effects of socially responsible investment. We find that the relation between socially responsible investment and the gender pay gap is strongest when there are two female directors within a firm. The effect of female directors on the relation comes from reduced information asymmetry between investors and firms.

Although there is an increasing demand for gender equality, most countries do not have disclosure requirements for gender equality. Information asymmetry works as an obstacle for investors in picking companies that treat gender equality better. A few countries, such as Denmark in 2006 and the United Kingdom in 2017, have implemented the disclosure requirements (Lagaras et al., 2022; Bennedsen et al., 2022). The United States attempted to implement a similar requirement, but the discussion halted in 2018. The findings of our papers imply that we need to require detailed information on gender equality for socially responsible investment to be effective.

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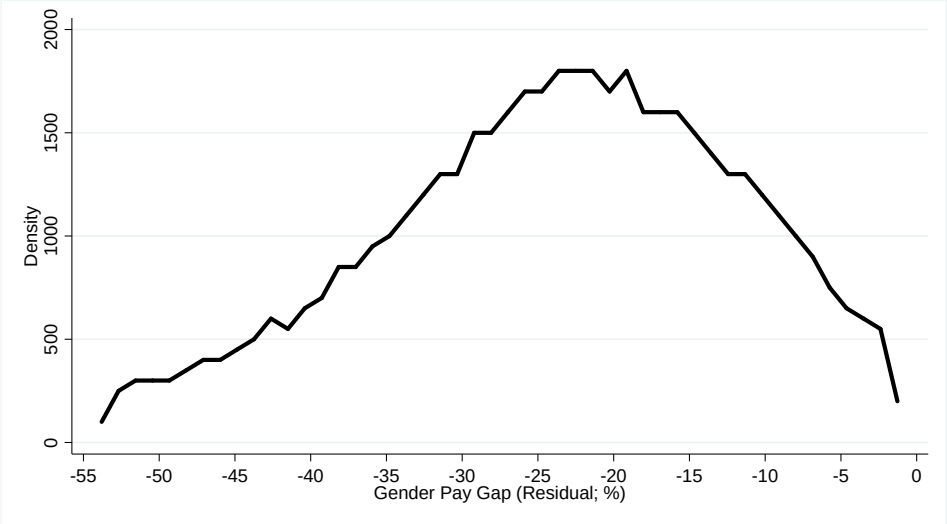
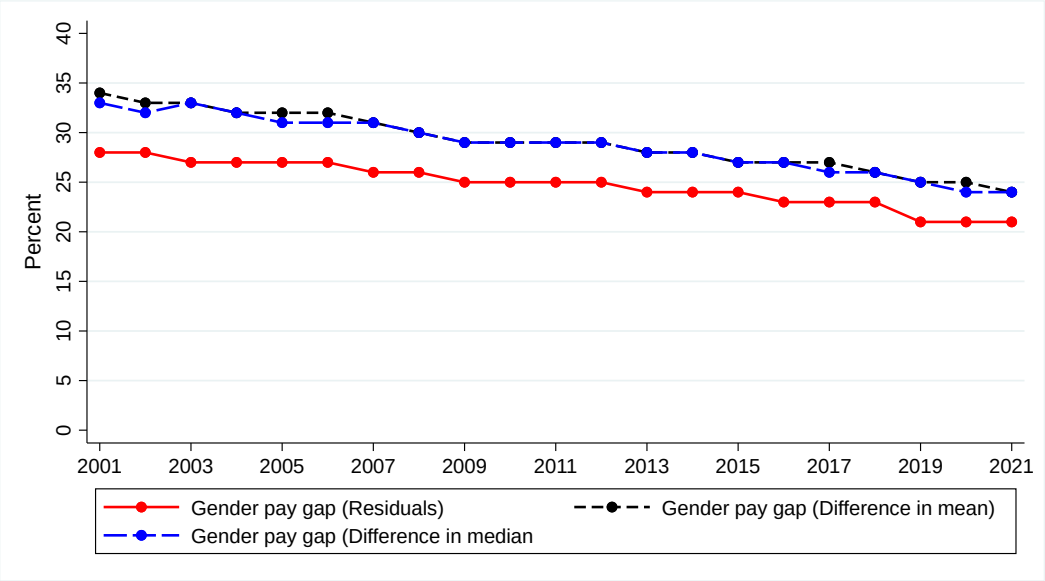


Figure 1. Kernel density plot of the gender pay gap

This figure presents the kernel density plot of the gender pay gap. We measure the gender pay gap by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. Section 2.2 provides details on the definition of the gender pay gap. To comply with the US Census disclosure requirements, we drop the top 5% and the bottom 5% of the firm-year observations. The sample firms consist of all companies at least 10 employees in Longitudinal Employer-Household Dynamics (LEHD) with valid financial data in Compustat from 2001 to 2021.

Panel A: Time trends of gender pay gaps



Panel B: Time trends of gender compositions of employees and directors

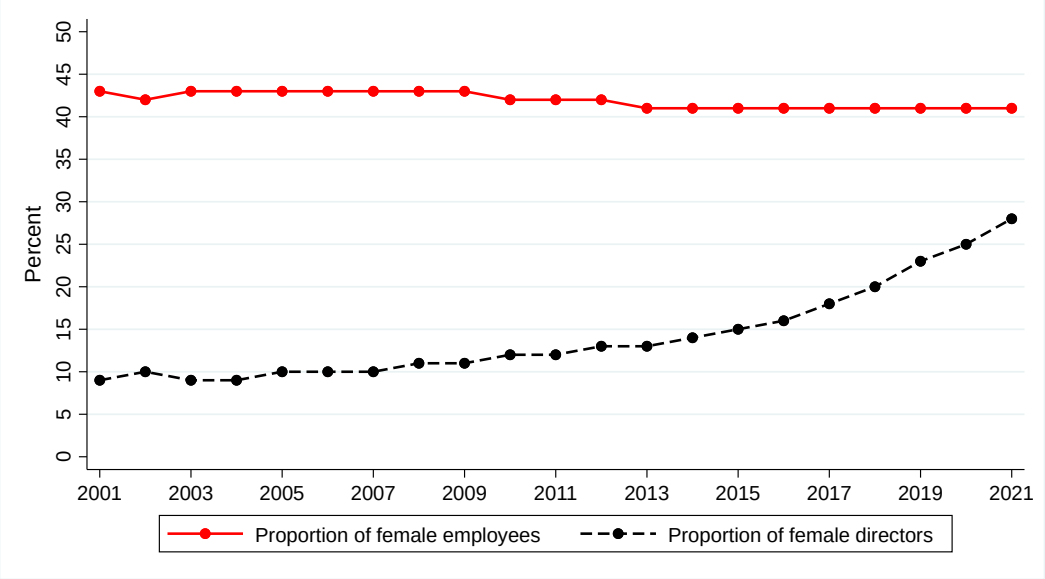


Figure 2. **Time trend of the within-firm gender gap and gender composition**
The figure shows the equally weighted average gender pay gap and female proportions of public companies from 2001 to 2021. Panel A shows the time trend of three measures of gender pay gaps. The residual measure is calculated by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. The other two measures compare the mean (median) male wages and female wages. Section 2.2 provides details on the definition of the three measures of the gender pay gap. Panel B shows the time trends of the number of female employees over all employees and the number of female directors over all directors. Prior to averaging, we winsorize numbers at the 1% and 99% levels.

Table 1. Summary Statistics

The table provides summary statistics for the gender pay gap. The sample firms consist of all companies with at least 10 employees in Longitudinal Employer-Household Dynamics (LEHD) with valid financial data in Compustat from 2001 to 2021. The gender pay gap is measured in three different ways. First, it is measured as a regression coefficient on female employee indicators within a firm-year. Section 2 provides additional details on the construction of the measure. The second measure is the mean of male wages minus the mean of female wages scaled by the mean of all employees' wages. The third measure is the median male wage minus the median female wage, scaled by a median of all employees. Definitions of other control variables are explained in Appendix Table A1. We winsorize ratios at the 1% and 99% levels. In the last column, we report the average characteristics of all Compustat companies from 2001 to 2021 for comparison.

	Sample firms			Compustat firms
	Mean	Std	Skewness	Mean
Gender pay gap	24.80	15.49	0.54	-
Gender pay gap (Difference in mean)	29.28	20.60	0.87	-
Gender pay gap (Difference in median)	28.83	25.48	1.29	-
Proportion of female employees	42.04	20.26	0.38	-
Proportion of female directors	13.89	12.57	0.60	12.43
Socially responsible investment	0.65	1.36	3.50	0.85
SRI by mutual funds' name	0.03	0.09	4.10	0.03
Institutional investment	67.97	29.02	-0.64	60.86
Non-socially responsible investment	67.28	28.75	-0.64	59.98
ln(Assets)	21.21	1.94	0.28	20.61
ln(Employments)	7.55	1.70	0.37	7.14
ln(Firm age)	3.24	0.50	-1.58	2.45
Profitability	7.00	10.40	-1.50	3.16
Book leverage	59.28	25.87	0.25	55.67
Number of female directors	1.21	1.17	0.90	1.10
Number of firm-years	54,000			58,564

Table 2. Gender pay gap and socially responsible investment

The table presents regression results for the gender pay gap on lagged socially responsible investment and firm characteristics. The dependent variable is the gender pay gap, measured in three ways. The benchmark measure is calculated by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. The other two measures compare the mean (median) male wages and female wages. Socially responsible investment of a firm is an aggregation of socially responsible institutional investors' holding in 13F divided by the number of shares outstanding. An institutional investor is classified as a socially responsible institutional investor if they increase the holding when the proportion of female directors increases for the previous five years. Alternatively, we use mutual fund names and classify a mutual fund as a socially responsible investor if the name of the fund includes words related to the social agenda. Section 2 provides details on the definition of the main dependent and independent variables. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	Gender pay gap	Gender pay gap	Gender pay gap	Gender pay gap	Gender pay gap (Mean)	Gender pay gap (Median)	Gender pay gap
Socially responsible	-0.481***	-0.446***	-0.073	-0.058	-0.004	-0.020	
Investment	(-7.49)	(-7.20)	(-1.43)	(-1.02)	(-0.06)	(-0.21)	
SRI by fund name							-0.557
							(-0.52)
ln(Assets)		1.537***	0.338***	0.366***	0.296*	0.457**	0.371***
		(10.99)	(2.88)	(2.92)	(1.77)	(2.35)	(2.95)
ln(Employment)		-1.518***	0.116	0.092	0.966***	-0.435*	0.093
		(-9.54)	(0.81)	(0.60)	(4.71)	(-1.77)	(0.61)
ln(Firm age)		1.137***	0.989***	0.927**	0.918*	1.259**	0.927**
		(3.00)	(2.65)	(2.33)	(1.79)	(1.98)	(2.33)
Profitability		-0.053***	-0.014	-0.019	-0.016	-0.056***	-0.019
		(-3.54)	(-1.19)	(-1.50)	(-0.99)	(-2.76)	(-1.49)
Leverage		0.019**	-0.028***	-0.030***	-0.045***	-0.037***	-0.030***
		(2.44)	(-4.86)	(-4.83)	(-5.65)	(-3.89)	(-4.83)
Constant	24.73***	-0.852	14.97***	14.92***	15.06***	20.39***	14.80***
	(110.10)	(-0.35)	(6.82)	(6.38)	(4.93)	(5.60)	(6.33)
SIC3-fixed	No	No	Yes	No	No	No	No
State-fixed	No	No	Yes	No	No	No	No
Year-fixed	No	No	Yes	No	No	No	No
SIC3*Year-fixed	No	No	No	Yes	Yes	Yes	Yes
State*Year-fixed	No	No	No	Yes	Yes	Yes	Yes
Adj. R-squared	0.00	0.04	0.37	0.35	0.40	0.37	0.35
N of firm-years	54,000	54,000	54,000	54,000	54,000	54,000	54,000

Table 3. Proportion of females and socially responsible investment

The table presents regression results for the proportion of females with lagged socially responsible investment and firm characteristics. The dependent variable is the number of female employees over total employment (Columns 1 to 3) or the number of female directors over all directors (Columns 4 to 6). Socially responsible investment of a firm is an aggregation of socially responsible institutional investors' holding in 13F divided by the number of shares outstanding. An institutional investor is classified as a socially responsible institutional investor if they increase the holding when the proportion of female directors increases for the previous five years. Alternatively, we use mutual fund names and classify a mutual fund as a socially responsible investor if the name of the fund includes words related to the social agenda. Section 2 provides details on the definition of socially responsible investment. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable	Proportion of female employees			Proportion of female directors		
Socially responsible	-0.421***	-0.089*	-0.096*	0.285***	0.116**	0.114**
investment	(-4.81)	(-1.81)	(-1.70)	(5.60)	(2.52)	(2.18)
ln(Assets)	0.076	-0.498***	-0.520***	1.812***	1.634***	1.606***
	(0.40)	(-3.89)	(-3.80)	(22.26)	(18.38)	(16.92)
ln(Employment)	-0.582**	(0.20)	-0.215	0.272***	0.409***	0.428***
	(-2.35)	(-1.33)	(-1.34)	(2.74)	(3.92)	(3.80)
ln(Firm age)	-1.476***	0.830**	0.853**	1.814***	-0.051	0.016
	(-2.98)	(2.41)	(2.36)	(6.63)	(-0.19)	(0.06)
Profitability	-0.060***	-0.001	-0.004	0.001	0.030***	0.026**
	(-3.28)	(-0.11)	(-0.30)	(0.10)	(2.95)	(2.31)
Leverage	0.166***	0.017***	0.016**	0.012**	0.019***	0.019***
	(15.65)	(2.65)	(2.25)	(2.13)	(3.59)	(3.20)
Constant	40.42***	50.45***	51.04***	-33.39***	-25.13***	-24.83***
	(12.62)	(22.80)	(21.73)	(-23.19)	(-15.07)	(-13.98)
SIC3-fixed	No	Yes	No	No	Yes	No
State-fixed	No	Yes	No	No	Yes	No
Year-fixed	No	Yes	No	No	Yes	No
SIC3*Year-fixed	No	No	Yes	Yes	No	Yes
State*Year-fixed	No	No	Yes	Yes	No	Yes
Adj. R-squared	0.05	0.7	0.69	0.12	0.33	0.31
N of firm-years	54,000	54,000	54,000	54,000	54,000	54,000

Table 4. Market valuations and employee-level and board-level gender equality

The table presents regression results for the market valuations on the same-year employee-level gender pay gap and the proportion of female directors. The dependent variable is the market value of assets over the book value of assets. The employee-level gender pay gap is calculated by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. The proportion of the female board of directors is the number of female directors over a total number of directors at the end of the fiscal year. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Dependent variable	Market Assets/Book Assets		
Gender pay gap	-0.139		-0.152*
	(-1.60)		(-1.74)
Proportion of female directors		0.593***	0.598***
		(5.36)	(5.40)
ln(Assets)	-5.806***	-6.707***	-6.773***
	(-5.08)	(-5.84)	(-5.88)
ln(Employments)	1.206	0.972	0.947
	(0.97)	(0.78)	(0.76)
ln(Firm age)	-41.57***	-41.45***	-41.59***
	(-12.26)	(-12.29)	(-12.31)
Profitability	3.247***	3.228***	3.232***
	(14.86)	(14.8)	(14.82)
Leverage	0.163**	0.148*	0.153*
	(2.07)	(1.88)	(1.94)
Constant	403.7***	420.4***	418.3***
	(19.28)	(19.88)	(19.85)
SIC3*Year-fixed	Yes	Yes	Yes
State*Year-fixed	Yes	Yes	Yes
Adj-R-squared	0.29	0.30	0.30
N	54,000	54,000	54,000

Table 5. Gender pay gap on interaction between female directors and socially responsible investment

The table presents regression results for the gender pay gap on lagged socially responsible investment, female directors, and firm characteristics. The dependent variable is the gender pay gap, measured in three ways, and the proportion of female employees over total employment. The residual measure is calculated by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. The other two measures compare the mean (median) male wages and female wages. "1(Number of female directors ≥ 1)" equals one if there is at least one female director at the end of the previous year. Socially responsible investment of a firm is an aggregation of socially responsible institutional investors' holding in 13F divided by the number of shares outstanding. An institutional investor is classified as a socially responsible institutional investor if they increase the holding when the proportion of female directors increases for the previous five years. Alternatively, we use mutual fund names and classify a mutual fund as a socially responsible investor if the name of the fund includes words related to the social agenda. Section 2 provides details on the definition of the main dependent and independent variables. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	Gender pay gap	Gender pay gap	Gender pay gap	Gender pay gap (Difference in mean)	Gender pay gap (Difference in median)	Gender pay gap	Proportion of female employees
[1] Socially Responsible investment	-0.200*	0.106	0.117	0.179	0.252		-0.147
	(-1.86)	(1.18)	(1.16)	(1.45)	(1.53)		(-1.62)
[2] 1(Number of female directors ≥ 1)	-2.907***	-0.531*	-0.437	-0.552	-0.133	-0.456	1.518***
	(-7.52)	(-1.70)	(-1.29)	(-1.27)	(-0.24)	(-1.37)	(5.02)
[1] * [2]	-0.333***	-0.271***	-0.266**	-0.278*	-0.416**		0.065
	(-2.62)	(-2.60)	(-2.28)	(-1.88)	(-2.17)		(0.61)
[3] SRI by fund name						5.703***	
						(3.57)	
[2] * [3]						-8.082***	
						(-4.07)	
ln(Assets)	1.792***	0.390***	0.411***	0.350**	0.484**	0.419***	-0.640***
	(12.41)	(3.26)	(3.21)	(2.06)	(2.43)	(3.27)	(-4.57)
ln(Employment)	-1.473***	0.132	0.106	0.984***	-0.426*	0.112	-0.251
	(-9.30)	(0.92)	(0.69)	(4.79)	(-1.73)	(0.73)	(-1.57)
ln(Firm age)	1.330***	1.002***	0.941**	0.933*	1.273**	0.948**	0.830**
	(3.51)	(2.69)	(2.36)	(1.82)	(2.00)	(2.38)	(2.31)
Profitability	-0.055***	-0.014	-0.019	-0.016	-0.056***	-0.019	-0.004
	(-3.77)	(-1.17)	(-1.50)	(-0.99)	(-2.76)	(-1.49)	(-0.30)
Leverage	0.021***	-0.027***	-0.030***	-0.045***	-0.036***	-0.030***	0.015**
	(2.78)	(-4.79)	(-4.77)	(-5.59)	(-3.85)	(-4.77)	(2.09)
Constant	-5.412**	14.03***	14.09***	14.07***	19.78***	13.87***	52.99***
	(-2.16)	(6.31)	(5.94)	(4.53)	(5.32)	(5.85)	(22.13)
SIC3-fixed	No	Yes	No	No	No	No	No
State-fixed	No	Yes	No	No	No	No	No
Year-fixed	No	Yes	No	No	No	No	No
SIC3*Year-fixed	No	No	Yes	Yes	Yes	Yes	Yes
State*Year-fixed	No	No	Yes	Yes	Yes	Yes	Yes
Adj. R-squared	0.04	0.04	0.04	0.40	0.37	0.35	0.69
N of firm-years	54,000	54,000	54,000	54,000	54,000	54,000	54,000

Table 6. Alternative specifications determinants of the gender pay gap

The table presents regression results for the gender pay gap on lagged socially responsible investment, firm characteristics, and other determinants. The dependent variable is the gender pay gap, calculated by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. The socially responsible investment of a firm is an aggregation of socially responsible institutional investors' holding in 13F divided by the number of shares outstanding. An institutional investor is classified as a socially responsible institutional investor if they increase the holding when the proportion of female directors increases for the previous five years. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
Dependent variable		Gender pay gap		
[1] Socially responsible investment	-0.063		0.112	0.121
	(-1.09)		(1.07)	(1.17)
[2] 1(Number of female directors ≥ 1)		0.364	-0.437	-0.438
		(0.52)	(-1.29)	(-1.29)
[3] Non-Socially responsible investment		0.022***		
		(2.84)		
[4] 1(Female CEO)	-1.748**			
	(-2.26)			
[1] * [4]	0.139			
	(0.42)			
[2] * [3]		-0.016*		
		(-1.78)		
[1] * [2]			-0.265**	-0.264**
			(-2.27)	(-2.20)
[1] * [1(Year ≥ 2012)]				-0.012
				(-0.11)
[Socially responsible investment]^2			0.00	
			(0.31)	
ln(Assets)	0.371***	0.347***	0.411***	0.411***
	(2.96)	(2.66)	(3.21)	(3.21)
ln(Employment)	0.092	0.095	0.106	0.106
	(0.60)	(0.61)	(0.69)	(0.69)
ln(Firm age)	0.947**	0.984**	0.941**	0.941**
	(2.38)	(2.47)	(2.36)	(2.36)
Profitability	-0.019	-0.025*	-0.019	-0.019
	(-1.48)	(-1.90)	(-1.50)	(-1.50)
Leverage	-0.030***	-0.028***	-0.030***	-0.030***
	(-4.82)	(-4.52)	(-4.77)	(-4.77)
Constant	14.81***	14.06***	14.09***	14.09***
	(6.34)	(5.95)	(5.94)	(5.94)
SIC3*Year-fixed	Yes	Yes	Yes	Yes
State*Year-fixed	Yes	Yes	Yes	Yes
Adj. R-squared	0.35	0.35	0.35	0.35
N of firm-years	54,000	54,000	54,000	54,000

Table 7. The relation between the gender gap and female director composition by the number of female directors
The table presents regression results for the gender pay gap on lagged socially responsible investment, female directors, and firm characteristics. The dependent variable is the gender pay gap, calculated by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. 1(Number of female directors = *n*) equals one if there are *n* female directors at the end of the previous year. The socially responsible investment of a firm is an aggregation of socially responsible institutional investors' holding in 13F divided by the number of shares outstanding. An institutional investor is classified as a socially responsible institutional investor if they increase the holding when the proportion of female directors increases for the previous five years. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Dependent variable		Gender pay gap	
[1] Socially responsible	-0.211**	0.101	0.114
investment	(-1.96)	(1.12)	(1.13)
1(Female Director = 1)	-1.696***	-0.423	-0.35
	(-4.38)	(-1.34)	(-1.02)
[1]* 1(Female Director = 1)	-0.285**	-0.187	-0.180
	(-2.00)	(-1.59)	(-1.38)
1(Female Directors = 2)	-3.711***	-0.649*	-0.447
	(-7.76)	(-1.66)	(-1.04)
[1]* 1(Female Directors = 2)	-0.460***	-0.382***	-0.404***
	(-2.92)	(-3.00)	(-2.81)
1(Female Directors ≥ 3)	-7.162***	-1.527***	-1.485***
	(-10.65)	(-2.93)	(-2.61)
[1] * 1(Female Directors ≥ 3)	-0.124	-0.294*	-0.273
	(-0.62)	(-1.91)	(-1.53)
ln(Assets)	2.083***	0.452***	0.470***
	(14.16)	(3.74)	(3.63)
ln(Employment)	-1.477***	0.136	0.110
	(-9.41)	(0.95)	(0.72)
ln(Firm age)	1.577***	1.013***	0.953**
	(4.16)	(2.72)	(2.39)
Profitability	-0.056***	-0.013	-0.019
	(-3.87)	(-1.15)	(-1.48)
Leverage	0.023***	-0.027***	-0.029***
	(3.06)	(-4.68)	(-4.67)
Constant	-12.09***	12.76***	12.86***
	(-4.64)	(5.68)	(5.36)
SIC3-fixed	No	Yes	No
State-fixed	No	Yes	No
Year-fixed	No	Yes	No
SIC3*Year-fixed	No	No	Yes
State*Year-fixed	No	No	Yes
Adj. R-squared	0.06	0.37	0.35
N of firm-years	54,000	54,000	54,000

Table 8. Discussion of gender-related words during earnings calls

The table presents regression results for the proportion of gender-related words in earnings calls on female directors and firm characteristics. The dependent variable is the number of gender-related words spoken by firm insiders over the total number of words by insiders during earnings calls. We use two groups of wordlist. The first wordlist includes "gender," "female," "DEI," and "D&I," and the second wordlist is the first word list plus "diversity," "inclusion," and "equality." "1(Number of female directors ≥ 1)" equals one if there is at least one female director at the end of the previous year. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dep. variable	Proportion of List 1 gender-related words				Proportion of List 2 gender-related words			
1(Number of female	0.765***	0.407***	0.185**	0.155**	13.007***	6.000***	2.306***	2.392***
directors ≥ 1)	(7.78)	(4.08)	(2.00)	(2.11)	(19.49)	(9.31)	(3.89)	(3.54)
ln(Assets)		0.372***	0.387***	0.412***		0.536**	0.790**	0.776*
		(8.62)	(5.89)	(5.38)		(1.97)	(2.00)	(1.72)
ln(Employment)		-0.135***	-0.083	-0.105		3.515***	3.800***	3.799***
		(-3.17)	(-1.24)	(-1.34)		(13.02)	(9.03)	(7.84)
ln(Firm age)		-0.088*	-0.106**	-0.109*		0.033	-0.384	-0.394
		(-1.78)	(-2.06)	(-1.84)		(0.11)	(-1.32)	(-1.19)
Profitability		0.282	0.602**	0.447		10.395***	11.610***	10.485***
		(1.18)	(2.40)	(1.54)		(7.27)	(7.94)	(6.39)
Leverage		-0.003	-0.003	-0.003		0.054***	0.024*	0.024*
		(-1.30)	(-1.44)	(-1.19)		(4.28)	(1.91)	(1.69)
Constant	2.943***	-2.254***	-3.046***	-3.299***	31.316***	-29.648***	-33.864***	-33.609***
	(37.94)	(-3.53)	(-4.00)	(-3.76)	(59.90)	(-7.23)	(-7.58)	(-6.65)
SIC3-fixed	No	Yes	No	No	Yes	No	No	Yes
State-fixed	No	Yes	No	No	Yes	No	No	Yes
Year-fixed	No	Yes	No	No	Yes	No	No	Yes
SIC3*Year-fixed	No	No	Yes	No	No	Yes	No	No
State*Year-fixed	No	No	Yes	No	No	Yes	No	No
Adj. R-squared	0.01	0.02	0.10	0.07	0.05	0.14	0.32	0.31
N of firm-years	38,153	38,153	38,153	38,153	38,153	38,153	38,153	38,153

Appendix

Table A1. Construction methods for variables
The table explains how we construct dependent variables and control variables. All fractional numbers are expressed in percentage terms.

Variables	Description
Gender pay gap	Following Goldin et al. (2017), we regress the logarithm of annual wage on an indicator variable for gender along with other individual characteristics within each firm-year. We exclude top seven high income individuals within a firm-year in order to exclude managers or directors. After the regression, we take the negative value of $\beta_{f,t}$ to measure the gender pay gap. $$\log(AnnualWage)_{i,t} = \beta_{f,t} * 1(Female)_i + Race_i + BirthCountry_i + Education_i + Age_{i,t} + Experience_{i,t}$$ "1(Female)" takes one if the gender at birth is female and zero if it is male. "Race" is four indicator variables, taking value of one if an individual at birth is an Asian-American, African-American, Hispanic-American, or other races, respectively. White-American individuals are benchmark. "BirthCountry" is an indicator variable which takes one if an individual is born outside of the United States. "Education" is three indicator variables, taking value of one if an individual's final education level is high school or equivalent, college or associate degree, and Bachelor's degree or advanced degree, respectively. Individuals without high school diploma are benchmark. "Age" is a integer variable of biological age. "Experience" is number of years since an individual had first valid salary data in each firm.
Gender pay gap (Difference in mean)	Within a firm-year, we subtract female employees' average wage from male employees' average salary and divide it by all employees' average wage. We exclude top seven high income individuals within a firm-year in order to exclude managers or directors.
Gender pay gap (Difference in median)	Within a firm-year, we subtract female employees' median wage from male employees' median salary and divide it by all employees' median wage. We exclude top seven high income individuals within a firm-year in order to exclude managers or directors.
Socially responsible investment	Sensitivity of institutional holdings in response to changes in the female composition of the board of directors with the previous 5 years' data. Specifically, for each institutional investor i, we analyze how they change their holding depending on the number of female directors for the previous 5 years. $$\Delta \frac{Investment}{AssetsUnderManagement}_{i,f,t-n} = \gamma_{i,t} * \Delta \frac{FemaleDirectors}{AllDirectors}_{f,t-n} + Controls_{f,t-n}$$ Controls include log of assets, net income over assets, and investment over assets. If an institution-year has positive $\gamma_{i,t}$ and it is statistically significant, we classify the institution as socially responsible in year t. For a firm's socially responsible investment, we aggregate holdings by socially responsible institutions and scale them by the number of shares outstanding.
SRI by fund name	As outlined in van der Beck (2023), we mutual fund names with words related to social concerns to classify them as socially responsible institutional investors. The word list we use is: social, impact, fair, gender, screen, ethical, conscious, SRI, ESG, and CSR. The last three words are case-sensitive, and other words are not case-sensitive. We aggregate socially responsible mutual funds' holdings within a firm-year and divide them by the number of shares outstanding
Institutional investment	Sum of shares held by all institutional investors observed in Thomson/Refinitiv 13F Holding data divided by the number of shares outstanding.
Non-socially responsible investment	Institutional investment minus socially responsible investment
Market-to-book assets	Market value of assets (PRCC_F*CSHO) as a percentage of book value of assets (AT) at the fiscal year end.

1(Number of female directors ≥ 1)	Indicator variable equal to one if there is at least female director at the end of the fiscal year
ln(Assets)	Log of real assets (AT*1'000'000).
ln(Employment)	Log of total employments. It includes entire states, not only the LEHD states.
ln(Firm age)	Log of years since first CRSP listing
Profitability	Net Income (NI) divided by lagged assets (AT)
Leverage	Book leverage defined as total debt (DLC + DLTT) divided by total assets (AT)
Proportion of List *n* gender-related words	Proportion of gender-related words spoken by managers during the earnings call. We use managers' scripts, including both introductory presentations and answers to the Q&A sessions. We aggregate quarterly earnings calls within a year to increase the power of the analysis. We use two groups of word lists. The first wordlist includes "gender," "female," "DEI," and "D&I," and the second wordlist is the first-word list plus "diversity," "inclusion," and "equality." We separate the word list because the latter three may include topics unrelated to gender equality of employees (such as "product diversity"). The dependent variable of this analysis is the proportion of gender equality words, which is the number of words related to gender equality mentioned in earnings calls throughout the year, scaled by the total number of words spoken in earnings calls.

Table A2. Additional evidence on determinants of gender pay gap.

The table presents regression results for the gender pay gap on lagged investor and firm characteristics. The dependent variable is the gender pay gap, calculated by regressing the log annual salary on female indicators and other individual characteristics within a firm-year. Institutional investment is the aggregate holding of any institutional investors on Thomson/Refinitiv 13F holding data divided by total outstanding shares. Non-socially responsible investment is calculated by subtracting socially responsible investment from the total institutional investment. An institutional investor is classified as a socially responsible institutional investor if they increase the holding when the proportion of female directors increases for the previous five years. Section 2 provides details on the definition of the main dependent and independent variables. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Dependent variable		Gender pay gap	
Institutional investment	0.011**		
	(1.98)		
Non-socially responsible investment		0.012**	
		(2.02)	
Proportion of female board of directors			-0.028**
			(-2.06)
log(Total employments)	0.078	0.078	0.103
	(0.51)	(0.50)	(0.67)
log(Firm age)	0.960**	0.961**	0.929**
	(2.41)	(2.41)	(2.33)
Net income/Assets	-0.024*	-0.024*	-0.019
	(-1.83)	(-1.84)	(-1.46)
log(Assets)	0.308**	0.307**	0.410***
	(2.41)	(2.40)	(3.23)
Book Leverage	-0.029***	-0.029***	-0.030***
	(-4.66)	(-4.66)	(-4.75)
Constant	15.31***	15.33***	14.21***
	(6.56)	(6.57)	(6.04)
SIC3*Year-fixed	Yes	Yes	Yes
State*Year-fixed	Yes	Yes	Yes
Adj-R-squared	0.35	0.35	0.35
N	54,000	54,000	54,000

Table A3. Proportion of females and socially responsible investment and female directors

The table presents regression results for the proportion of females with lagged socially responsible investment, female directors, and firm characteristics. The dependent variable is the number of female employees over total employment. Socially responsible investment of a firm is an aggregation of socially responsible institutional investors' holding in Thomson/Refinitiv 13F Holding data divided by the number of shares outstanding. An institutional investor is classified as a socially responsible institutional investor if they increase the holding when the proportion of female directors increases for the previous five years. "1(Number of female directors ≥ 1)" equals one if there is at least one female director at the end of the previous year. Definitions of all control variables can be found in Appendix Table A1. All continuous variables are winsorized at the 1% and 99% levels. Numbers in parentheses represent t-statistics, with standard errors clustered by firm. Significance levels are denoted by *, **, and ***, indicating significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
Dependent variable	Proportion of female employees		
[1] Socially responsible investment	-0.521***	-0.116	-0.147
	(-3.82)	(-1.45)	(-1.62)
[2] 1(Number of	4.219***	1.587***	1.518***
female directors ≥ 1)	(7.81)	(5.62)	(5.02)
[1] * [2]	0.089	0.028	0.065
	(0.53)	(0.30)	(0.61)
ln(Assets)	-0.278	-0.623***	-0.640***
	(-1.41)	(-4.75)	(-4.57)
ln(Employments)	-0.641***	-0.237	(0.251
	(-2.61)	(-1.58)	(-1.57)
ln(Firm age)	-1.740***	0.811**	0.830**
	(-3.52)	(2.37)	(2.31)
Profitability	-0.057***	-0.002	-0.004
	(-3.12)	(-0.14)	(-0.30)
Leverage	0.163***	0.016**	0.015**
	(15.41)	(2.49)	(2.09)
Constant	46.6***	52.46***	52.99***
	(14.02)	(23.25)	(22.13)
SIC3-fixed	No	Yes	No
State-fixed	No	Yes	No
Year-fixed	No	Yes	No
SIC3*Year-fixed	No	No	Yes
State*Year-fixed	No	No	Yes
Adj-R-squared	0.06	0.7	0.69
N	54,000	54,000	54,000